



Enhanced multimodal recommendation systems through reviews integration

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Abstract

Multimodal recommendation systems aim to capture diverse user preferences through data such as text and images, offering more personalized recommendation services. Accurately grasping user preferences can enhance the precision of recommendations and augment the user experience. Existing multimodal graph learning models enhance item representations by capturing the latent attribute relationships between items. However, multimodal features often suffer from modality missing issues, which hinder the model's ability to fully capture the connections between item attributes. Meanwhile, review data, as a form of textual information, not only provide descriptions of item attributes but also directly reflect user preferences. However, directly incorporating review data into item representations may introduce additional noise, potentially affecting the model's performance. Therefore, we propose the Personalized Multi-Preference Recommender (PMPR) model, which integrates reviews with multimodal data to extract multifaceted user preferences, enhancing the personalization of recommendations. Specifically, we designed a heterogeneous graph learning module based on reviews and a homogeneous graph learning module that combines the review features with multimodal features to capture users' diverse preferences. Considering the varying informational content of reviews, PMPR processes each review individually and utilizes user IDs to generate review attention vectors for aggregating the review features to reduce review noise. Finally, we integrate the Top-K method for recommendation. We compare common review processing methods with PMPR's approach to validate its effectiveness. In comparative analyses across five publicly available datasets, our enhanced model consistently demonstrated superior performance when benchmarked against seven widely recognized models. The results

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indicate a noteworthy improvement over the current state-of-the-art (SOTA) model, ranging from 2.76% to 22.52% in terms of average performance.

Keywords Multimodal recommendation · Multi-preference · Graph structure learning

1 Introduction

With the development of the information age, multimodal information has also been widely used in various tasks [1, 2]. In e-commerce, the proliferation of multimodal information (such as text and images) related to items has enabled users to gain a more comprehensive understanding of their favored item. Simultaneously, the reviews left by users regarding their purchase experiences provide valuable decision-making references for potential buyers. Past users with similar preferences have often made similar choices in the future [3–5]. Therefore, identifying and utilizing this multimodal information and review to delineate user preferences accurately is crucial for enhancing the precision of recommendation systems.

Early work [6] focused on mining user preference information by utilizing historical user–item interaction data within the collaborative filtering (CF) framework to provide personalized user recommendations. However, this method resulted in relatively simplistic user preferences and was negatively impacted by the sparsity of interaction data, leading to sub-optimal performance.

In addressing the limitations identified above, subsequent work in recommendation systems introduced richer information sources for predicting user preferences. Initially, many recommendation methods sought to forecast user preferences by gathering their rating data. However, this approach often needed more interaction data, yielding moderate effectiveness. Later, recommendation systems began to leverage user-generated content, such as textual reviews, to mitigate this sparsity issue. As supplementary information to ratings, these reviews elucidate why users rate items in specific ways and contain insights about user preferences and item pros and cons. For instance, the DeepCoNN [7] model employs convolutional networks to train on user and item review documents, thereby learning representations for both. Similarly, NARRE [8] processes each review to extract the most helpful information, subsequently using attention vectors to aggregate these reviews to form representations of users and items. While innovative, these methods heavily rely on the richness of user reviews. Given the inherent limitations of textual information, such approaches face constraints in their expressive capabilities.

Recent recommender systems enhance item feature representations by incorporating multimodal information, which effectively alleviates cold-start and data sparsity issues. Several studies [9–11] combine multimodal features as auxiliary information with the CF framework to enhance item representation. Nevertheless, the incorporation of multimodal information directly into the user–item interaction graph, lacking in-depth exploration, results in a constrained expressive capability of the model. Recent advancements in graph neural networks (GNNs) have found successful applications across various domains [12–14], notably in recommendation systems where GNNs have been extensively utilized. For instance, MMGCN [15] employs modal features of items to construct modal-specific user–item interaction graphs, facilitating the learning of user preferences within each modality. Similarly, GRCN [16] uses multimodal features to identify and pry potential false positive noise edges to enhance graph structure. Subsequent research has aimed at improving the representation of user preferences by incorporating auxiliary graph structures that capture interrelation-

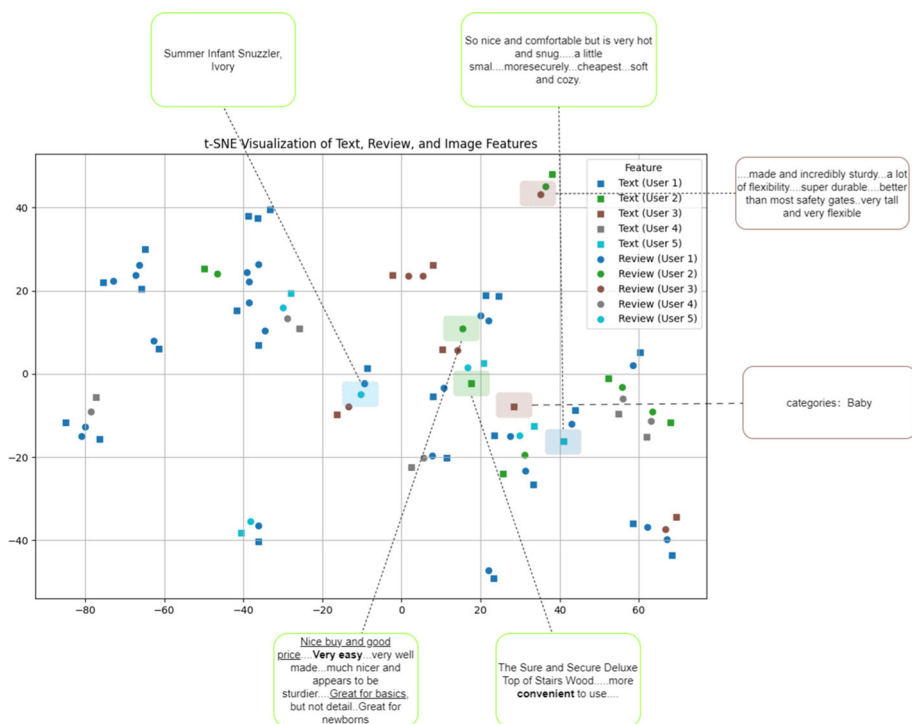


Fig. 1 Randomly selected five users and performed t-SNE dimensionality reduction on the text features and review features of the products they interacted with. In the resulting visualization, squares represent text features, while circles represent review features. Different colors indicate the distinct users

ships between users or items. Examples include DualGNN [17] and LATTICE [18], which introduce user–user and item–item graphs alongside the user–item graph, leading to more comprehensive user preference representations. Additionally, approaches such as FREEDOM [19] employ edge pruning techniques tailored to node degrees to refine the user–item graph, achieving state-of-the-art performance with strategies like freezing the item–item graph during training. Therefore, the evolution from traditional CF-based approaches to incorporating multimodal features and graph neural networks has significantly advanced recommendation systems’ capability to capture and accurately represent user preferences.

However, most of the aforementioned methods fail to adequately address the issue of modality missing in multimodal features. Through t-SNE dimensionality reduction analysis of textual and review features in the dataset (as shown in Fig. 1), the data can be categorized into the following types: one type includes textual features that only contain category information, lacking product descriptions or with empty descriptions, representing cases of missing text; another type consists of brief product descriptions that only mention the product’s shape or color, indicating incomplete textual descriptions; the final type includes detailed product descriptions that cover various functionalities but mainly focus on objective attributes, without addressing aspects such as cost-effectiveness or availability. In these cases, incorporating review data can effectively supplement the deficiencies of textual features and provide more preference information for the recommendation system.

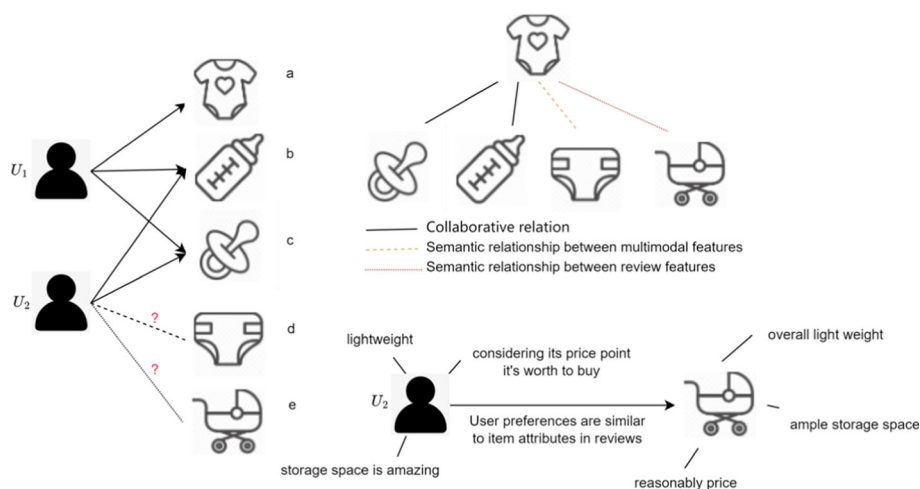


Fig. 2 Depicts an example of personalized recommendations integrating multimodal data and reviews. We contend that relying solely on multimodal data is insufficient, and integrating reviews contributes to providing users with more personalized recommendations

For example, as shown in Fig. 2, traditional collaborative filtering methods typically assume that user U_1 and user U_2 share similar preferences and recommend item a to user U_2 . By integrating existing multimodal techniques, the recommendation system can evaluate the relevance of multimodal features associated with items and recommend item d to the user. However, since item e lacks similarity with past user interactions in terms of textual or image features, existing methods are unable to recommend item e to user U_2 . By leveraging review data, it is possible to uncover the intrinsic attribute similarities between item a and item e and further identify that the preference attributes of user U_2 align with the intrinsic attributes of item e, enabling the recommendation of item e. Therefore, combining review data with multimodal features not only compensates for the lack of potential connections due to modality missing but also facilitates multi-perspective modeling of user preferences, enhancing the personalization of recommendations. Additionally, missing image links and the monotony of image content in the dataset also limit the expressiveness of image features, which, like textual features, primarily reflect partial objective attributes.

In recent years, contrastive learning has achieved numerous successful cases in recommendation systems. This method models data representations by minimizing the distance between similar data points while maximizing the distance between dissimilar ones. For example, the Review-enhanced Hierarchical Contrastive Learning model ReHCL [20] utilizes cross-view graph contrastive learning to enhance node representations, extract useful review information, and match rating and review representations through contrastive learning. Meanwhile, MCCL [21] points out that existing contrastive learning methods have not fully utilized the rich semantic information present in multimodal data, developing a multi-level contrastive learning framework that self-supervises to leverage both intra-modal and inter-modal semantic information. Overall, these models improve the ability of recommendation systems to handle review data through various methods and complexities.

Inspired by recent successes in multi-view learning [22, 23], we propose a personalized multi-preference recommendation model that integrates review data with multimodal features, aiming to provide a more comprehensive modeling of user preferences. Specifically,

PMPR utilizes a review-based processing method to preprocess reviews and extract features from reviews for denoising. Subsequently, it combines review features with multimodal features to construct an item–item graph, enabling the learning of more comprehensive latent attribute relationships between items. Additionally, a review-based user–item heterogeneous graph is designed to uncover user preferences within the reviews. Finally, the learned representations are integrated, and a Top-K method is employed for recommendations.

In summary, our work makes the following four main contributions:

- We propose a novel multimodal recommendation model that combines reviews and multimodal features of items to explore user preferences in depth, facilitating a more comprehensive learning of representations for both users and items.
- We find that the attributes in reviews differ from the textual and visual representations of items, and there is a modal absence in the items. To address this, we design a denoising strategy to extract review features. By combining multimodal features with review features, we delve deeper into user preferences, thereby enhancing the representation capabilities of both users and items.
- Extensive experimental validation across three publicly available datasets was undertaken to assess the efficacy of the proposed methodologies.

2 Related work

2.1 Reviews-based recommendation

Early recommendation algorithms primarily relied on collaborative filtering but faced performance limitations due to the sparsity of interaction data. Subsequent efforts incorporated reviews to mitigate this issue, dividing the approach into document-level and review-level recommendation methods. Document-based methods merge reviews into a single document to learn representations, such as the DeepCoNN [7] model, which employs two convolutional networks, and the CARL [24] model, which utilizes attention mechanisms to extract comprehensive information by learning word correlations. Review-level methods process each review individually for finer representation, as seen in the NARRE [8] model, which applies convolutional networks and attention vectors for aggregation, and the NRPA [25] model, which focuses on the significance of each word in a review for denoising. The AHOR [26] model employs heterogeneous graphs to capture aspect-aware relationships, enriching the learning of high-order representations for users and items through multi-hop connections, and selectively focusing on relevant features using attention mechanisms. However, these methods directly encode reviews for recommendation without deeply understanding the complex relationships between user behavior and item characteristics.

2.2 Multimodal content-based recommendation

Recent advancements integrate item modality data such as text and images to enhance recommendation accuracy. For instance, the VBPR [9] method uses CNNs to extract image features combined with IDs for recommendations, while the VECF [10] method segments images using reviews as a supervisory signal. With the significant progress of Graph Neural Networks (GNN), many studies have utilized GNNs to handle graph structures in recommendation systems. The MMGCN [15] model is designed to create bipartite user–item graphs specific to each modality, facilitating the learning process of user attention across various

modalities. Building on this, DualGNN [17] introduces a user co-occurrence graph, modeling inter-user connections. At the same time, GRCN [16] employs a graph refinement layer for denoising recommendations, reducing the weight of pseudo-interaction edges. LATTICE [18] establishes item–item relation graphs tailored to individual modalities, thereby effectively augmenting connections between items and mitigating cold-start challenges. FREEDOM [19] employs a degree-sensitive edge pruning approach to refine the user–item interaction graph, highlighting the importance of maintaining a static item–item graph during training. Additionally, examples of employing self-supervised learning in multimodal recommendation tasks are observed, as exemplified by SLMRec [27] and BM3 [28], aligning modal features for learning user and item representations and reducing model complexity. However, these methods overlook the issue of modality missing in multimodal features, which leads to insufficient exploration of user preferences.

2.3 Recommendation methods combining multimodal information and review information

In recent years, many studies have incorporated review information into multimodal recommendation systems to enhance node representation. McAuley et al. [29] provides a comprehensive overview of review-based recommendation methods. IHGCL [30] employs Intent-Intent Contrast and Intent-Interaction Contrast, along with a Bottlenecked Autoencoder, to fully explore the latent fine-grained intents underlying meta-paths. The MEGCF [31] model extracts semantic entities from multimodal features and review features to perform denoising, while using sentiment information from reviews as edge weights. The ReHCL [20] model is designed as a hierarchical contrastive learning framework to fully exploit features within reviews: it captures the structural similarity between themes and semantic views on one hand, and matches the relationship between reviews and ratings on the other. Subsequently, many studies began to extract aspect-level features from reviews for denoising. For instance, COSMOS [32] identified that different modalities express knowledge with varying ranges and details, leading to the design of a collaborative learning model to integrate different types of knowledge features. SERMON [33] pointed out that previous methods had not adequately fused the implicit feedback reflected in user–item interactions and review features, thus proposing a multimodal contrastive learning model to deeply explore multimodal features and review information while enhancing model interpretability.

Inspired by these methods, we identified challenges in integrating multimodal features with review features. Models like FREEDOM [19] utilize multimodal features to explore the relationships between items, thereby enhancing item node representations. However, multimodal features can suffer from text deficiencies and may not encompass sufficient attribute information. Review information can directly reflect user preferences and contains attribute features absent in multimodal data. To address this, we propose supplementing the existing multimodal item graph by constructing an isomorphic graph, while also incorporating review features into the user–item interaction graph to fully mine user preference information. This approach effectively integrates the implicit information from multimodal features and review features, helping to alleviate data sparsity issues. Additionally, when review information is limited, new items can still be recommended based on multimodal features, thereby mitigating the cold-start problem.

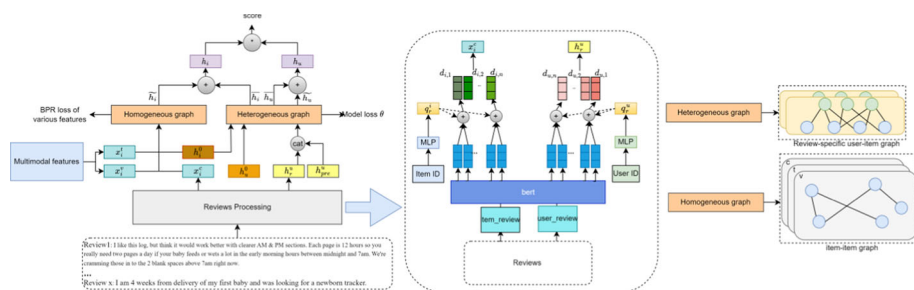


Fig. 3 The diagram outlines the core components of PMPR. Initially, reviews undergo preprocessing, then extracting representations for both items and users from the enhanced item–item isomorphic graphs and user–item heterogeneous graphs, utilizing the processed information. Ultimately, these representations are integrated with a Top-K strategy to facilitate recommendations

3 Methodology

In this section, we will outline the essential details of the PMPR model, with Fig. 3 depicting the internal structure of our model. It is primarily composed of four parts: (1) A dual-attention mechanism is employed to denoise review information and extract latent representations of users and items from the reviews; (2) Homogeneous graph learning, which combines review features with multimodal features, is used to capture more comprehensive relationships between items; (3) Heterogeneous graph learning enhanced by reviews is leveraged to explore user preferences embedded in the reviews; (4) A prediction module calculates scores based on the final representations of users and items, followed by a Top-K strategy for recommendation.

3.1 Preliminary

Given a user–item interaction graph $G = (u, i, e | u \in U, i \in I, e \in \varepsilon)$, an interaction is represented as $e_{u,i} = 1$ if it exists; otherwise, $e_{u,i} = 0$, where $e_{u,i} \in E$, $u \in U$, and $i \in I$. The notation R^{UI} signifies the statement made by the user U on item I . In cases where a review is absent, we employ a vector adhering to a normal distribution for substitution. The terms h_u^0 and h_i^0 denote the ID embeddings for users and items, respectively. An item's multimodal data is represented as $M = \{t, v, c\}$, corresponding to text, images, and reviews. In this study, we consider reviews as a third training modality due to the subjective nature of preference information embedded in reviews.

3.2 The feature learning

It is crucial to fully utilize the review information, as modeling based on documents may introduce noise into the generated representations of users and items, thus interfering with experimental results. Therefore, we preprocess the collected review on a per-review basis and ultimately aggregate reviews into personalized attention vectors based on the content of the reviews and the corresponding user IDs. Notably, our review only retains the review associated with user–item interactions in the training set while excluding reviews related to user–item interactions present in the validation and test sets.

Text feature extraction We concatenate the item's title, category, brand, and description as j_i , and then use sentence-transformers [34] to extract 384-dimensional text features. This is illustrated by Equation (1):

$$W_{u,i} = \text{Sentence_BERT}(j_i) \quad (1)$$

Visual feature extraction The visual features are directly provided by Amazon, specifically using 4096-dimensional visual features extracted with a pretrained convolutional neural network [35, 36]. This is illustrated by Equation (2):

$$W_{u,i} = \text{Deep_CNN}(\text{image}_i) \quad (2)$$

Review feature extraction Given review information $r_{u,i}$, in our experiment, the approach to processing reviews differs from previous methods. We employ the same sentence encoder to process item text information and extract features from the review information. The rationale behind this approach is to recognize that review information itself is textual data, and utilizing the same processing method places it in the same vector space as the textual modality, facilitating integration in subsequent steps. This is illustrated by Equation (3):

$$W_{u,i} = \text{SentenceTransformer}(r_{u,i}) \quad (3)$$

For each user, there exists a set of interacted items, along with a corresponding set of reviews. We concatenate the transformed features of each review to form the review information feature for that user. Subsequently, we convert the review information features of the user into a matrix $M_u = [W_{u,i_1}, W_{u,i_2}, \dots, W_{u,i_t}]$, in which t represents the number of items interacted with by the user.

After acquiring review information features attributed to each user, we introduce a user-specific attention mechanism to emphasize relevant reviews for individual users. This approach acknowledges that not all reviews hold equal informativeness, with each review bearing varying degrees of significance. It is important to note that some items may have empty review sets. To ensure the uninterrupted progress of the experiments, we assign a randomly generated vector conforming to a normal distribution to the empty reviews of this item.

Generate user-specific attention vectors for reviews: Based on the user ID embedding U_{id} , we employ a multilayer perceptron (MLP) to process U_{id} , thus generating personalized review-level attention vectors for each user, as illustrated by Equation (4).

$$q_r^u = \text{Leaky_ReLU}(W_1 * U_{id} + b_1) \quad (4)$$

Here, Leaky_ReLU denotes the activation function, W_1 represents the weight matrix, and b_1 denotes the bias term.

Integration of the review features For the generated user review information feature matrix M_u , we utilize review-level attention vectors to emphasize informative reviews while attempting to downplay irrelevant ones. Specifically, we compute the weight of the j review for the i user as depicted in Eqs. (5) and (6).

$$u_j = q_r^u A_1 W_{u,i_j} \quad (5)$$

$$\beta_j = \text{Softmax}(u_j) \quad (6)$$

In this context, A_1 represents the attention matrix, while q_r^u denotes the user's review-level attention vector. Based on the generated review weights β_j , we integrate the user's reviews

to derive the latent preference feature representation of user u , as delineated in Eq. (7):

$$h_r^u = \sum_{j=1}^t \beta_j W_{u,i_j} \quad (7)$$

Similarly, we can also obtain the latent preference feature representation x_i^c for the item.

3.3 Learning of homogeneous graphs

In multimodal features, the potential relationships between items have garnered attention. Unlike previous approaches that use multimodal features as edge information, this study employs multimodal features to learn the item graph structure and directly models item–item relationships using item ID embeddings. This method allows for the implicit modeling of item–item relationships through item–user–item relationships. However, relying solely on multimodal features is insufficient; therefore, we introduce review information to complement multimodal features, aiming to more comprehensively model the relationships of items. Here, we consider review information as a third modality of item features rather than simply adding it as textual information to existing text features. To achieve this, we employ the k-nearest neighbors (kNN) method to process multimodal features. Specifically, we calculate the similarity between the multimodal features of each item and the review features, and we aggregate these edges to construct a more comprehensive item–item graph. Based on the cosine similarity of the original features between items x_i and x_{i_1} , we compute an edge e_{i,i_1} , as shown in equation (8):

$$e_{i,i_1}^{m'} = \frac{(x_i^{m'})^T (x_{i_1}^{m'})}{\|x_i^{m'}\| \|x_{i_1}^{m'}\|} \quad (8)$$

Here, $m' \in M$ denotes the modal features of an item, where M represents the set of modalities. The graph constructed in this manner is a fully connected network. Links with extremely low scores are deemed negligible, allowing us to retain only the edges corresponding to the top- k similarity scores for each item [18]. Furthermore, we aim to learn the fundamental relational structure between items from the modal features, so we convert the constructed graph into an unweighted one, and the resulting unweighted graph is normalized. The specific details of this process are outlined in Eq. (9):

$$e_{i,i_1}^{m'} = \begin{cases} 1 & \text{if } e_{i,i_1}^{m'} \in \text{top} - K(e_{i,i_1}^{m'}) \\ 0 & \text{otherwise} \end{cases} \quad (9)$$

Having obtained the item–item graphs for each modality, we combine them by performing weighted summation based on the importance score $\rho_{m'}$. This importance score represents the contribution of each graph form in M toward the final item graph and can be learned through a parameterized function. Formally, this is represented as $e_{i,i_1} = \sum_{m' \in M} \rho_{m'} e_{i,i_1}^{m'}$.

We apply graph convolution to the acquired item graphs, wherein during the training phase, we freeze the item-to-item graph. This helps improve training efficiency [18], as delineated in Equation (10):

$$\widetilde{h_i^{(l+1)}} = \sum_{i_1 \in N_i} e_{i,i_1} \widetilde{h_{i_1}^{(l)}} \quad (10)$$

where, $\widetilde{h_i^{(0)}} = h_i^0$, h_i^0 represents the item ID embedding.

3.4 Learning of heterogeneous graphs

The characteristics of reviews contain direct preference information from users. To better extract user preferences from review information, we construct a user–item heterogeneous graph focused on review features, denoted as G_{re} , based on the original user–item interaction graph. To comprehensively consider users' static features and dynamic behaviors, we initialize a user representation and connect it with the review feature representation (h_{pre}^u), inputting it into the graph G_{re} . We use LightGCN to perform information propagation and aggregation on the graph G_{re} . Through graph convolution, we learn the representations of user and item nodes, as shown in Eqs. (11) and (12):

$$u_r^{(l+1)} = \sum_{i \in N_u} \frac{1}{\sqrt{|N_u|}\sqrt{|N_i|}} i_r^{(l)} \quad (11)$$

$$i_r^{(l+1)} = \sum_{u \in N_i} \frac{1}{\sqrt{|N_u|}\sqrt{|N_i|}} h_{pre}^u \quad (12)$$

where, N_u and N_i represent the sets of neighbor nodes for u and i in graph G_{re} . The features learned at each layer are normalized using a symmetric normalization $\frac{1}{\sqrt{|N_u|}\sqrt{|N_i|}}$. After propagation through L layers, we aggregate the representations obtained at each GCN layer by summation to derive the final user feature representation, as illustrated in Eq. (13):

$$\widetilde{h}_u = \sum_{l=0}^L u_r^{(l)} \quad (13)$$

In addition, we apply a degree-sensitive random pruning method for noise reduction on the original user–item interaction graph, which is validated in [37]. High-degree nodes tend to exhibit over-smoothing during training, so we use random pruning to reduce edges by a certain ratio during training. We define the pruned interaction graph as G_{ui} and use LightGCN to propagate and aggregate information on the graph. We input the user and item IDs into the graph, and after information aggregation, we obtain $\widetilde{h}_u$ and $\widetilde{h}_i$.

3.5 Feature fusion

We apply convolution to the original user–item interaction graph to obtain $\widetilde{h}_u$ and $\widetilde{h}_i$. The learned representation $\widetilde{h}$ encapsulates a combination of static and dynamic user behaviors. Here, we decompose $\widetilde{h}_u$ into $\widetilde{h}_{u'}$ and $\widetilde{h}_{u''}$ and then integrate these components at a γ ratio. In our experiments, γ can be learned from parametric functions. The learned features of users and items are then integrated, as detailed in Eqs. (14) and (15):

$$Z_u = \widetilde{h}_u + \gamma \widetilde{h}_{u'} + (1 - \gamma) \widetilde{h}_{u''} \quad (14)$$

$$Z_i = \widetilde{h}_i + \widetilde{h}_i \quad (15)$$

3.6 Optimization

To fully mine the information in the features, we use multilayer perceptron (MLP) to process the multimodal features and review features. The specific process is shown in Eq. (16).

$$h_i^{m'} = x_i^{m'} W_{m'} + b_{m'} \quad (16)$$

In this context, $W_{m'}$ and $b_{m'}$ respectively denote the weight matrix and bias within the Multilayer Perceptron (MLP).

To optimize the parameters within the PMPR framework, we employ the Pairwise Bayesian Personalized Ranking (BPR) loss. The primary objective of this approach is to ensure that the scores of positive user–item pairs surpass those of antagonistic user–item pairs. We incorporate the derived latent item features into the loss function, treating them as integral components. The definition of the loss function L is presented as Eqs. (17) and (18).

$$L = \sum_{(u,i,j) \in R} (-\log_{\sigma}(h_u^T h_i - h_u^T h_j) + \lambda L_{rec}) \quad (17)$$

$$L_{rec} = \sum_{m' \in M'} -\log_{\sigma}(h_u^T h_i^{m'} - h_u^T h_j^{m'}) + \| \theta \|_2 \quad (18)$$

In the given context, R represents the set of training instances. Here, σ denotes the sigmoid activation function, λ is a hyperparameter, and θ signifies the model parameters.

3.7 Top-K recommendation

To generate item recommendations, interaction scores between users and candidate items are predicted. These scores are then used to rank the candidate items, with the top-K items selected as user recommendations. The calculation methodology for interaction scores is detailed in Eq. (19).

$$score = Z_u^T Z_i \quad (19)$$

When scores are high, it implies greater user preference for the item. In our approach, we solely utilize embeddings of user and item IDs for prediction. Incorporating multimodal features of the predicted item did not enhance performance.

4 Experiments

4.1 Experiments settings

Datasets We have selected five datasets from Amazon, categorized under Baby, Sports, Clothing, Beauty and Tianmao [38]. Each dataset encompasses two types of data: item information descriptions and user rating records. For the item information, we concatenate the item name, category, and description to form a textual representation. This text is then processed using a sentence transformer, resulting in a 384-dimensional sentence embedding to serve as the text feature. For visual data, we utilize Amazon-provided 4096-dimensional image features that have been pre-extracted and published. In each dataset, we retained users and items with at least five interactions, ensuring that every user or item has a minimum of five interactions. This methodology has been frequently employed in previous studies.

Table 1 Data set statistics

Dataset	User	Item	Interactions	Sparsity (%)
Baby	19445	7050	160792	99.88
Sports	35598	18357	296337	99.95
Clothing	39387	23033	278677	99.97
Beauty	22363	12101	198502	99.92
Tianmao	1704	1642	11033	99.60

Additionally, we use user review from the user rating records as the review feature. This data is also processed with a sentence transformer, yielding another 384-dimensional sentence embedding for our experiments. Table 1 compiles and summarizes the dataset details. The total number of reviews corresponds to the number of interactions. Data sparsity is the ratio of interactions to the item of the total number of users and items.

Evaluation metrics In our experimental setup, we split the interaction data in an 8:1:1 ratio, wherein the review was exclusively utilized from the training subset of these interactions. In addition, we adopted the recognized evaluation metrics in recommender systems, namely top-K strategies: Recall@K and NDCG@K for evaluation, here referred to simply as R@K and N@K. We conducted experiments utilizing K values of 10 and 20, computing average metrics across all users within the test dataset. Adopting established empirical methodologies, our training included integrating a negative sampling strategy. To gauge the accuracy of our recommendations, we used a full-ranking protocol to compare different evaluation metrics. The above concisely summarizes the strategies and evaluation methods employed during our training process.

Hyperparameter settings Our proposed model was implemented using the PyTorch framework, setting the embedding dimension for users and items to 64. Each batch size was fixed at 2048, and the evaluations were conducted on a GPU with 32GB memory on an RTX 3060Ti. We used grid search to identify the optimal hyperparameters: we set the learning rate options to $\{0, 0.001, 0.0001, 0.00001\}$ and regularization terms to 0, 0.001, 0.0001, 0.00001. The optimal parameters were a learning rate 0.001 and a regularization term 0.001. We set the layer numbers for heterogeneous and homogenous graphs at 2 and 1, respectively. Based on experience, the visual feature ratio ρ_v was set at 0.1, while ρ_t and ρ_c were determined through grid search. The ρ_c values were set at $\{0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1\}$, with the optimal ρ_c parameter being 0.1. Experimental validation led to setting the drop ratio for user-item graphs in the Baby and Clothing datasets at 0.8 and the Sports dataset at 0.9. We set the maximum epochs to 1000 and employed an early stopping strategy.

4.2 Baseline systems

To validate the efficacy of PMPR, we conducted a comparative analysis against two groups of cutting-edge baseline methods. BPR and LightGCN were categorized as general recommendation models, while VBPR, GRCN, DualGNN, SLMRec, LATTICE, and FREEDOM were classified as multimodal recommendation models.

- BPR [6] optimizes personalized item ranking directly, employing traditional methods like MF or KNN for recommendation optimization.

- LightGCN [13] simplifies the structure of Graph Convolutional Networks (GCN) by eliminating feature transformation and nonlinear activation components, preserving solely the neighborhood aggregation module.
- VBPR [9] integrates visual features extracted from item into matrix factorization. We have combined both visual and textual aspects of item for learning item representations.
- GRCN [16] purifies the user–item interaction graph by removing false positive edges, then uses GCN on the purified interaction graph for information propagation and aggregation to learn representations of item and users.
- DualGNN [17] based on the user–item interaction graph, introduces a user co-occurrence graph to enhance user representations.
- SLMRec [27] introduces self-supervised learning in multimodal recommendation tasks, employing data augmentation on individual item in various ways, followed by contrastive learning to extract additional supervisory signals for exploring latent relationships between item modalities.
- LATTICE [18] uses each modality feature to construct item–item graphs, and then aggregates all the modal graphs to form the final potential item–item graph. The final representations of users and item are obtained through graph convolution on this and user–item graphs.
- FREEDOM [19] refines the LATTICE model by introducing a comparable item–item graph, with the additional step of freezing the graph pre-training. It further integrates a degree-sensitive edge pruning technique to eliminate a specific proportion of edges within the user–item graph based on node degree, thereby achieving a denoising effect. This significantly reduces space and improves training efficiency compared to the LATTICE model. We have enhanced user and item representations using review information on this basis.

5 Analysis

5.1 Performance comparison

We divide the state-of-the-art models into general and multimodal recommendation models and compare and analyze the recommendation performance with PMPR; the specific results are shown in Tables 2 and 3. Our observations and analyses are as follows:

1. Graph-based recommendation models, such as LightGCN, outperform traditional models like BPR. Because user–item interaction graphs are typically sparse, graph-based models are more adept at handling sparse data. Additionally, graph-based models can leverage the graph's propagation and information aggregation mechanisms to more effectively unearth the implicit features and complex relationships between users and items. The above experimental results also confirm the efficacy of using graphs with convolution operations to model historical interactions.
2. PMPR consistently outperforms all baseline models regarding Recall and NDCG on all datasets. On average, PMPR exceeds LATTICE by 22.52% and surpasses the FREEDOM model by 2.76% across all datasets. Regarding performance enhancement, the increase in NDCG is more significant than that in Recall, indicating that PMPR further amplifies the existing user preferences for items, advancing the recommendation ranking of items in the test set. However, in the Sports and Clothing datasets, the improvement in NDCG is almost on par with Recall. This may be due to more interactions than the

Baby dataset, resulting in a relatively lower proportion of useful review information, further diluting the initially helpful reviews. The improvement involves enhancing the representation of item–item homographs and user–item hetero-graphs through review information. Reviews yield direct reflections of user preferences and opinions about the item and reveal more details and features, such as quality and applicable scenarios. In the item–item homograph, we enrich the graph’s edge representation by capturing the fundamental relational structure between items as reflected in the reviews. In the user–item heterograph, we build a user-specific heterogeneous graph, thoroughly mining user preferences from the reviews. Hence, review information can provide rich user feedback and item characteristics, aiding our model in better understanding user needs and item properties, thereby constructing a more comprehensive graph for learning representations of users and items. The experimental findings indicate the effectiveness of our proposed model in integrating review data into multimodal recommendations. This integration enriches graph information and optimizes user and item representations.

3. Models incorporating modal features of items demonstrate superior performance compared to conventional Collaborative Filtering (CF) models. For instance, VBPR, which integrates visual features extracted from items into matrix factorization, outperforms BPR across all datasets. The most advanced models are currently based on the multimodal features of the item. Compared to a single modality, multimodal recommendation approaches utilize a variety of data types, which have complementary and enriching relationships with each other. This cross-modal complementarity can compensate for the limitations of a single modality, enhancing the accuracy and personalization of recommendation systems. Examples include multimodal models like LATTICE, SLMRec, GRCN, and FREEDOM, which outperform other methods in terms of performance. However, these multimodal methods often overlook the latent value embedded in review information. Our experiments enhance the representation of users and items by learning personalized characterizations from reviews, addressing this oversight.
4. Compared to the Baby, Sports, and Clothing datasets, the Beauty dataset exhibits a more severe issue with missing text. PMPR shows a significant improvement over FREEDOM on the Beauty dataset, indicating that utilizing review information to supplement missing text is highly effective. Additionally, the Tianmao dataset, after processing, is smaller in size, has fewer reviews, and lower quality, yet PMPR still outperforms FREEDOM across all metrics, demonstrating its robustness.

5.2 Represents similarity analysis

The PMPR model aims to mine users’ preferences in greater detail. We randomly sampled five users (represented in different colors) and all the items they interacted with from the Baby dataset. We used t-distributed stochastic neighbor embedding (t-SNE) to visualize the representations of users and items. As illustrated in Figs. 4 and 5, it is observed that within FREEDOM, the similarity between the majority of users and their genuinely selected items’ representations is lower than in PMPR. This observation aligns with the expected outcomes post-integration of reviews. This discrepancy is attributed to FREEDOM’s negligence of users’ intrinsic preferences in reviews. Therefore, relying solely on multimodal features fails to effectively simulate users’ personalized preferences. In contrast, PMPR efficiently encapsulates these personalized preferences.

Table 2 Performance evaluation was conducted through comparative analysis with a range of baseline methods focused on Recall, Map, Precision, F1 and NDCG metrics

Dataset	Metric	BPR	LightGCN	VBPR	GRCN	DualIGNN	SLMRec	LATTICE	FREEDOM	PMPR	improv(%)
Baby	R@10	0.0357	0.0479	0.0423	0.0532	0.0513	0.0521	0.0545	0.0637	0.0663	21.65%/4.08%
	R@20	0.0575	0.0754	0.0663	0.0824	0.0803	0.0772	0.0846	0.0994	0.1015	20.0%/2.11%
	N@10	0.0192	0.0257	0.0223	0.0282	0.0278	0.0289	0.0294	0.0332	0.0358	21.77%/7.83%
	N@20	0.0249	0.0328	0.0284	0.0358	0.0352	0.0354	0.0371	0.0424	0.0448	20.75%/5.66%
	Map@10	0.0159	0.0194	0.0178	0.0208	0.0192	0.0206	0.0209	0.0232	0.0257	22.97%/10.78%
	Map@20	0.0172	0.0211	0.0191	0.0228	0.0212	0.0226	0.0229	0.0256	0.0281	22.71%/9.77%
	P@10	0.0037	0.0053	0.0049	0.0059	0.0054	0.0057	0.006	0.007	0.0073	21.67%/4.29%
	P@20	0.0029	0.0044	0.0039	0.0046	0.0043	0.0044	0.0049	0.0055	0.0056	14.29%/1.82%
	F1@10	0.0067	0.0095	0.0087	0.0106	0.0097	0.0102	0.0108	0.0126	0.0131	21.3%/3.97%
	F1@20	0.0055	0.0083	0.0073	0.0089	0.0081	0.0083	0.0092	0.0104	0.0106	15.22%/1.92%
Sports	R@10	0.0432	0.0569	0.0558	0.0599	0.0588	0.0663	0.0620	0.0717	0.0727	17.26%/1.39%
	R@20	0.0653	0.0864	0.0856	0.0919	0.0899	0.0990	0.0953	0.1089	0.1113	16.47%/1.93%
	N@10	0.0241	0.0311	0.0307	0.0330	0.0324	0.0365	0.0335	0.0385	0.0392	17.01%/1.82%
	N@20	0.0298	0.0387	0.0384	0.0413	0.0404	0.0450	0.0421	0.0481	0.0490	16.39%/1.87%
	Map@10	0.0169	0.0221	0.0194	0.0232	0.0233	0.0273	0.024	0.0273	0.0279	16.25%/2.2%
	Map@20	0.0182	0.0231	0.0215	0.0253	0.0253	0.0295	0.0266	0.0299	0.0306	15.04%/2.34%
	P@10	0.0045	0.0061	0.0055	0.0068	0.0067	0.0074	0.0072	0.0078	0.0079	9.72%/1.28%
	P@20	0.0034	0.0048	0.0043	0.0052	0.0049	0.0055	0.0056	0.006	0.0062	10.71%/3.33%
	F1@10	0.0081	0.011	0.01	0.0122	0.012	0.0133	0.0129	0.014	0.0142	10.08%/1.43%
	F1@20	0.0064	0.009	0.0081	0.0098	0.0092	0.0104	0.0105	0.0113	0.0117	11.43%/3.54%

Table 2 continued

Dataset	Metric	BPR	LightGCN	VBPR	GRCN	DualGNN	SLMRec	LATTICE	FREEDOM	PMPR	improv(%)
Clothing	R@10	0.0206	0.0361	0.0281	0.0421	0.0452	0.0442	0.0492	0.0629	0.0640	30.08%/1.75%
	R@20	0.0303	0.0544	0.0415	0.0657	0.0675	0.0659	0.0733	<i>0.0941</i>	0.0956	30.42%/1.59%
	N@10	0.0114	0.0197	0.0158	0.0224	0.0242	0.0241	0.0268	<i>0.0341</i>	0.0346	29.10%/1.47%
	N@20	0.0138	0.0243	0.0192	0.0284	0.0298	0.0296	0.0330	<i>0.0420</i>	0.0427	29.39%/1.67%
	Map@10	0.0074	0.0152	0.0143	0.016	0.0173	0.0181	0.0213	<i>0.0247</i>	0.0254	19.25%/2.83%
	Map@20	0.0091	0.0169	0.0155	0.0176	0.0188	0.0195	0.0224	<i>0.0268</i>	0.0276	23.21%/2.99%
	P@10	0.0027	0.0042	0.0038	0.0045	0.0047	0.0047	0.0051	<i>0.0064</i>	0.0066	29.41%/3.12%
	P@20	0.0024	0.0033	0.0031	0.0035	0.0035	0.0034	0.004	<i>0.0048</i>	0.005	25%/4.17%
	F1@10	0.0047	0.0075	0.0066	0.0081	0.0085	0.0085	0.0092	<i>0.0116</i>	0.0119	29.35%/2.59%
	F1@20	0.0044	0.0062	0.0057	0.0066	0.0066	0.0064	0.0075	<i>0.0091</i>	0.0095	26.67%/4.4%

'improv(%)' values compare PMPR to LATTICE and FREEDOM. The calculation is as follows: $21.65\% = (R@10_{PMPR} - R@10_{LATTICE}) / R@10_{LATTICE}$. The previously leading results are in italics, and our experimental findings are emphasized in bold. 'Improv' represents the ratio the PMPR improves over the best-performing baseline. It is pertinent to mention that all experimental results are derived within the framework outlined in MIMRec [25]

Table 3 Performance evaluation was conducted through comparative analysis with a range of baseline methods focused on Recall, Map, Precision, F1 and NDCG metrics

Dataset	Metric	BPR	LightGCN	VBPR	GRCN	DualGNN	SLMRec	LATTICE	FREEDOM	PMPR	improv(%)
Beauty	R@10	0.0534	0.0773	0.068	0.0871	0.0926	0.0971	0.0932	0.1047	0.1128	16.42%/3.63%
	R@20	0.0877	0.1143	0.0998	0.1262	0.1346	0.1369	0.1347	0.1497	0.1608	16.18%/4.54%
	N@10	0.0334	0.0447	0.0376	0.0482	0.0526	0.0547	0.0521	0.0583	0.0631	15.74%/3.43%
	N@20	0.0421	0.0537	0.046	0.0585	0.0637	0.0652	0.063	0.0702	0.0757	15.56%/3.7%
	Map@10	0.0234	0.0322	0.0268	0.0344	0.0385	0.0395	0.0378	0.0424	0.0459	15.34%/2.83%
	Map@20	0.0273	0.0342	0.0291	0.0373	0.0415	0.0425	0.0408	0.0456	0.0493	15.2%/3.07%
	P@10	0.0064	0.0092	0.0083	0.0106	0.0112	0.0119	0.011	0.0122	0.0132	15.45%/4.1%
	P@20	0.049	0.0072	0.0062	0.0078	0.0082	0.0085	0.0081	0.0089	0.0096	14.81%/4.49%
	F1@10	0.0114	0.0164	0.0148	0.0189	0.02	0.0212	0.0196	0.0218	0.0236	15.82%/4.13%
	F1@20	0.0092	0.0135	0.0116	0.0147	0.0154	0.016	0.0153	0.0168	0.0181	14.38%/4.17%
Tianmao	R@10	0.8124	0.8251	0.8467	0.8363	0.8476	0.8671	0.8596	0.8617	0.8694	1.14%/0.89%
	R@20	0.8413	0.8763	0.0.8846	0.8824	0.8882	0.8987	0.8984	0.8938	0.8995	0.12%/0.64%
	N@10	0.6193	0.6147	0.6277	0.6332	0.6246	0.6642	0.6536	0.5909	0.5915	−9.5%/0.1%
	N@20	0.6311	0.6231	0.6375	0.6453	0.6343	0.6729	0.6641	0.6093	0.6103	−8.1%/0.16%
	Map@10	0.5267	0.5431	0.5522	0.5623	0.5754	0.5928	0.5819	0.5535	0.5634	−3.8%/1.79%
	Map@20	0.5341	0.5531	0.555	0.5659	0.5832	0.5955	0.5848	0.5786	0.5864	0.27%/1.35%
	P@10	0.0751	0.0854	0.0896	0.0884	0.0863	0.0914	0.0903	0.0873	0.0889	−1.55%/1.83%
	P@20	0.0367	0.0457	0.0478	0.0477	0.0467	0.0487	0.0483	0.0482	0.0484	0.21%/0.41%
	F1@10	0.1374	0.1548	0.162	0.1598	0.1566	0.1653	0.1634	0.1585	0.1598	−2.2%/0.82%
	F1@20	0.0703	0.0868	0.0906	0.0905	0.0887	0.0923	0.0916	0.0914	0.0918	0.22%/0.44%

Note: ‘improv(%)’ values compare PMPR to LATTICE and FREEDOM. The calculation is as follows: $21.65\% = (R@10_{PMPR} - R@10_{LATTICE}) / R@10_{LATTICE}$. The previously leading results are italics, and our experimental findings are emphasized in bold. ‘Improv’ represents the ratio the PMPR improves over LATTICE and FREEDOM. It is pertinent to mention that all experimental results are derived within the framework outlined in MMRec [25]

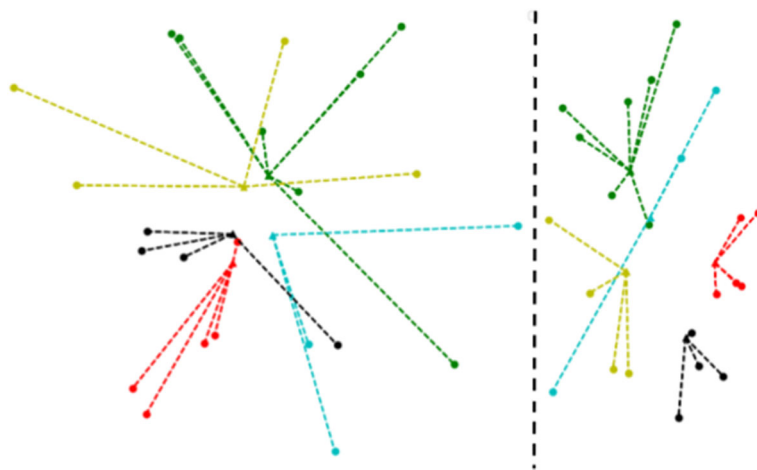


Fig. 4 Presents a visual comparison of the representations of users and items learned by PMPR and FREEDOM, as depicted through the t-SNE transformation. Each central dot of color represents a user, and connected to it are the items that the user prefers, with the length of the connecting lines indicating the degree of similarity

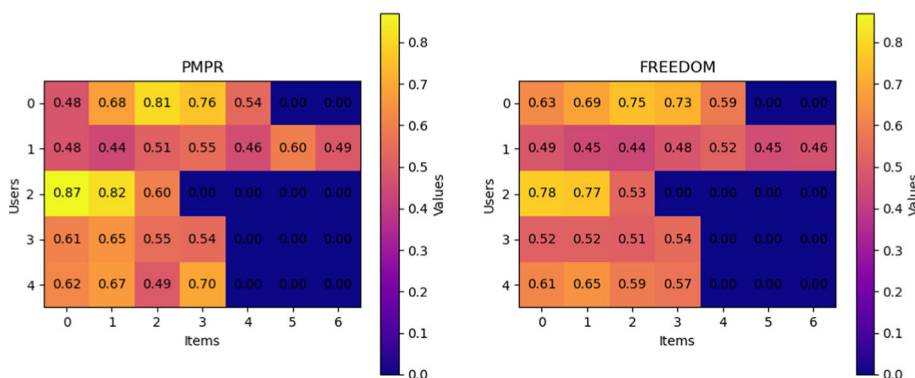


Fig. 5 The similarity is computed for the features of users and items learned by PMPR and FREEDOM, where the ordinate is the user number, and the abscissa is the item index

5.3 Ablation study

We designed the following variants based on each component in PMPR and how reviews are handled:

- PMPR_I retains only the item–item graph section that is enhanced using reviews
- PMPR_U retains only the user–item graph section that is enhanced using reviews
- PMPR_C uses reviews-based processing, Glove to process all words into word vectors, and CNN to extract features.
- PMPR_D uses a document-based processing approach to handle reviews and a sentence transformer to extract the review features.

Table 4 shows that on the Baby and Sports datasets, the PMPR_I and PMPR_U models outperform the FREEDOM model in terms of performance. However, their performance on the Clothing dataset could be much better. This underperformance may be attributed to a

Table 4 Experimental results of different modules of PMPR and different ways of processing reviews under three data sets

Model	Baby R@20	N@20	Sports R@20	N@20	Clothing R@20	N@20
LATTICE	0.0846	0.0371	0.0953	0.0421	0.0733	0.0330
FREEDOM	0.0994	0.0424	0.1089	0.0481	0.0941	0.0420
PMPR_D	0.0992	0.0438	0.1058	0.0464	0.0928	0.0413
PMPR_C	0.0997	0.0432	0.1093	0.0486	0.0926	0.0416
PMPR_U	0.1013	0.0445	0.1105	0.0489	0.0924	0.0415
PMPR_I	0.1002	0.0426	0.1106	0.0486	0.0925	0.0416
PMPR	0.1015	0.0448	0.1111	0.0491	0.0956	0.0427

potential imbalance in information when only reviews are used to enhance a single module while neglecting other aspects. On the Clothing dataset, the ratio of reviews to users is 7.075, lower than that on the Baby and Sports datasets, which stand at 8.269 and 8.3245, respectively. This indicates a lesser volume of reviews per user on the Clothing dataset and, consequently, less available helpful information, making the imbalance in information more pronounced.

5.4 Review validity

To corroborate the effectiveness of reviews in enhancing recommendations, we predominantly executed empirical validations in the following two dimensions:

Sample analysis of a single item: Initially, we extracted the genuinely liked item (i.e., the item with which users interacted in the test set) from the final recommendations of the state-of-the-art baseline models FREEDOM and PMPR. Figure 6 reveals that regardless of the review frequency, PMPR recommends more items than FREEDOM. Specifically, for user number 526, the items genuinely liked and recommended by FREEDOM was {1543}, while for PMPR, it was {1543, 5137, 2908, 674}. To further investigate, we deleted all related reviews for item {5137, 2908, 674} before training, leaving other review information unchanged. Subsequently, PMPR's recommended item changed to 1543, no longer recommending item {5137, 2908, 674}. This case analysis for individual items demonstrates the effectiveness of reviews in item recommendation.

Random drops: We investigated the impact of the frequency of user or item reviews on recommendation outcomes. A certain proportion of reviews for users and items was randomly dropped to verify the effectiveness of reviews in recommendations. It is important to note that if we do not categorize reviews or divide them into high-frequency and low-frequency groups, random dropping could remove some users' reviews altogether, necessitating random reassignment. Worse still, this approach could result in eliminating reviews from some users. Therefore, our experiment involved randomly dropping reviews for each user or item, specifically dropping 10, 30, and 50% of reviews from each. The results of this experiment are depicted in the following figure.

Figure 7 shows that for each dataset, the model's efficacy consistently declines as the proportion of dropped reviews increases. The difference between a 0% drop and a 10% drop is negligible. This can be attributed to the fact that with a relatively low drop rate, essential reviews are relatively removed, leading to minimal loss of information from the reviews.

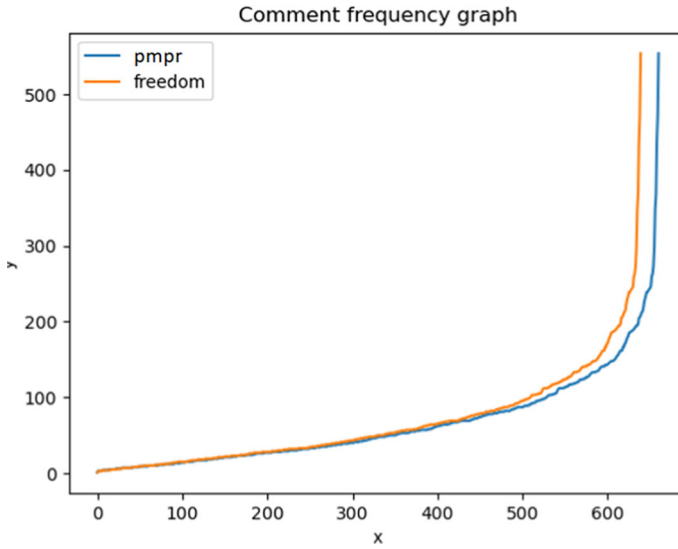


Fig. 6 Illustrates the relationship between the number of items ultimately recommended by FREEDOM and PMPR that users genuinely liked and the frequency of reviews. The x-axis corresponds to the number of recommended items, while the y-axis indicates the frequency of reviews

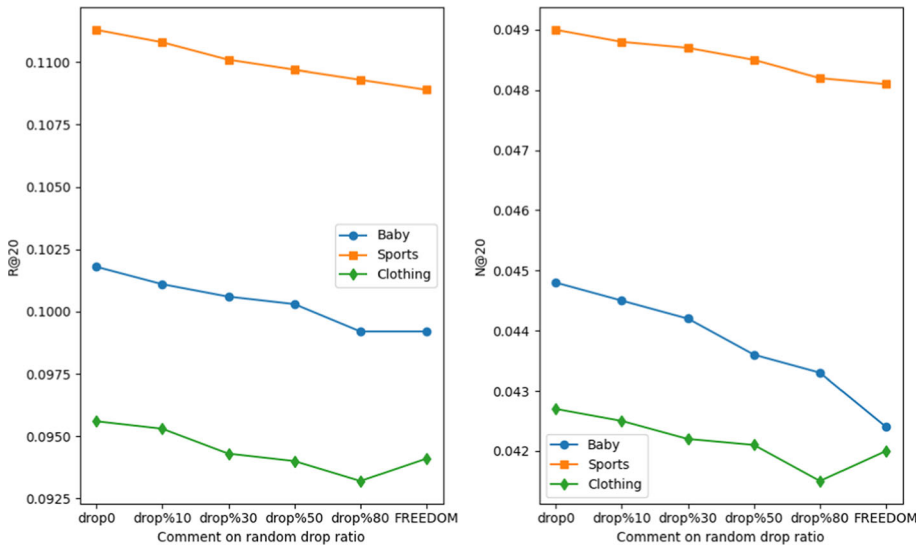
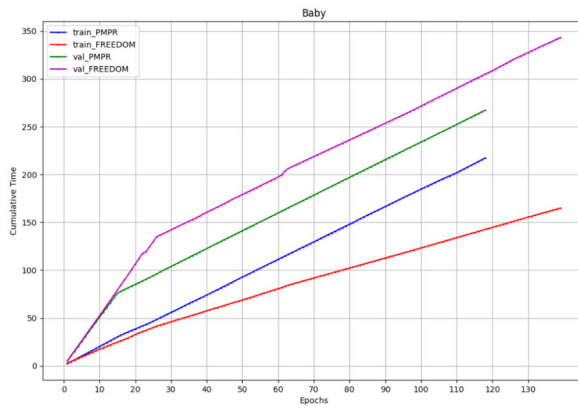
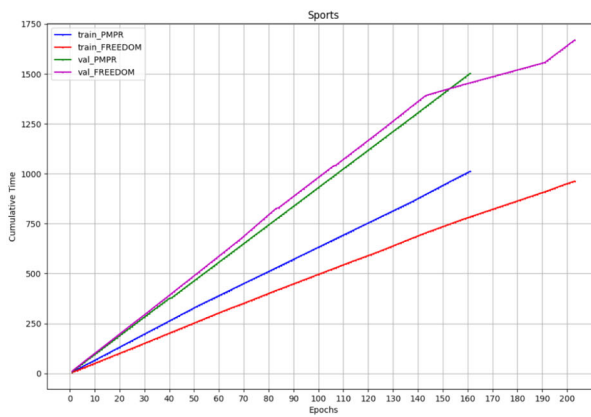


Fig. 7 Displays the outcomes of experiments conducted on each dataset after randomly dropping reviews. The x-axis depicts the proportion of dropped reviews, whereas the y-axis signifies the evaluation metrics

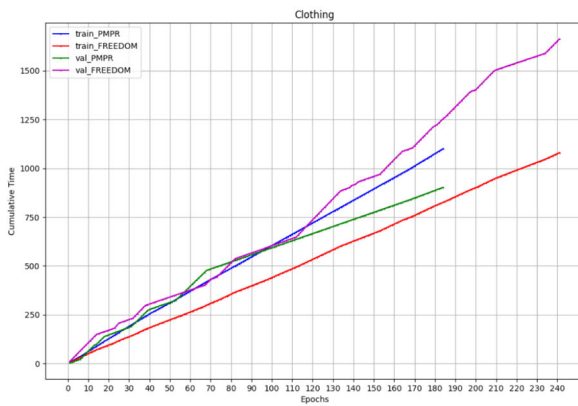
However, valuable reviews are likely eliminated as the drop ratio escalates, resulting in a noticeably diminished effect. Consequently, this demonstrates that the frequency of reviews has a discernible impact on the model's recommendation performance.



(a) Baby



(b) Sports



(c) Clothing

Fig. 8 We compare the training and prediction times of PMPR with FREEDOM, where the x-axis represents the number of training epochs and the y-axis shows the training time

Regarding the computational complexity of the proposed model, we conduct experiments comparing the PMPR model with the FREEDOM model in terms of training and prediction times, as shown in Fig. 8a–c.

During the training phase, we test the PMPR and FREEDOM models on different datasets. The results indicate that while PMPR converges more quickly on each dataset, the final training times are comparable to those of the FREEDOM model. However, in terms of prediction time, the PMPR model demonstrates superior performance, requiring significantly less time than FREEDOM. This suggests that, despite similar complexity during training, the PMPR model can make predictions more efficiently in practical applications, thereby enhancing overall efficiency.

5.5 Hyperparameter sensitivity study

Learning rate and loss weights: In the PMPR framework, we adjusted the learning rate, denoted as Learning_rate, from 0.01 to 0.0001 and the loss weight, labeled as λ , from 0.0 to $1e-05$. Figures 9(a), 9(b), and 9(c) display the performance of PMPR on three distinct datasets, Baby, Sports, and Clothing, under varying learning rates and loss weights. The performance was evaluated using the Recall@20 and NDCG@20 metrics. The results indicate that the optimum performance for both the Baby and Clothing datasets was achieved with the learning rate and loss weight set at 0.001. In contrast, for the Sports dataset, the best performance was observed at a learning rate of 0.001 and a loss weight of 0.0001. Compared to the learning rate, PMPR exhibited a lower sensitivity to changes in the loss weight; the variation in model performance was insignificant with the continuous alteration of the loss weight. However, an appropriately set loss weight can also significantly enhance the model's performance.

Fusion ratio when enhancing the item–item diagram based on review information: PMPR employs personalized representations of items extracted from the review to construct the item–item graph. We investigated the impact of the fusion ratio ρ_c on the performance of PMPR. As shown in Fig. 10, $\rho_c = 0$ value indicates no integration of the fundamental structure between items derived from review information. At the same time, $\rho_c = 1$ signifies no integration of the basic structure between items extracted from textual features. Figure 10 illustrates the performance of PMPR on three evaluation datasets in terms of Recall@20 and NDCG@20 metrics. The optimal performance was observed at $\rho_c = 0.1$, and performance gradually declined with increasing ρ_c value. This suggests that enhancing the item–item graph representation with review information is effective, as it encapsulates fundamental attributes of the item that are not reflected in the modal information of the item.

The ratio of text to image in merging item–item homogeneous graphs: As shown in Fig. 11, as the ratio of image features increases, R@20 and N@20 gradually decrease, reaching optimal performance at a ratio of 0.1. Therefore, we have set the fusion ratio of the image feature isomorphic graph and the text feature isomorphic graph to 0.1. We can infer that textual information offers richer content than visual information when constructing an effective item–item graph.

The fusion ratio of user features: As shown in Fig. 12, the parameters have a relatively minor impact on model performance in the Baby and Sports datasets, but the N@20 results are significant in the Clothing dataset. Specifically, in the Baby dataset, PMPR performs best when $\gamma=0.5$; in the Sports dataset, PMPR is optimal when $\gamma=0.3$; and in the Clothing dataset, PMPR achieves the best performance when $\gamma=0.6$. The results indicate that the importance of the two user groups is fundamentally similar, so combining them yields the best results.

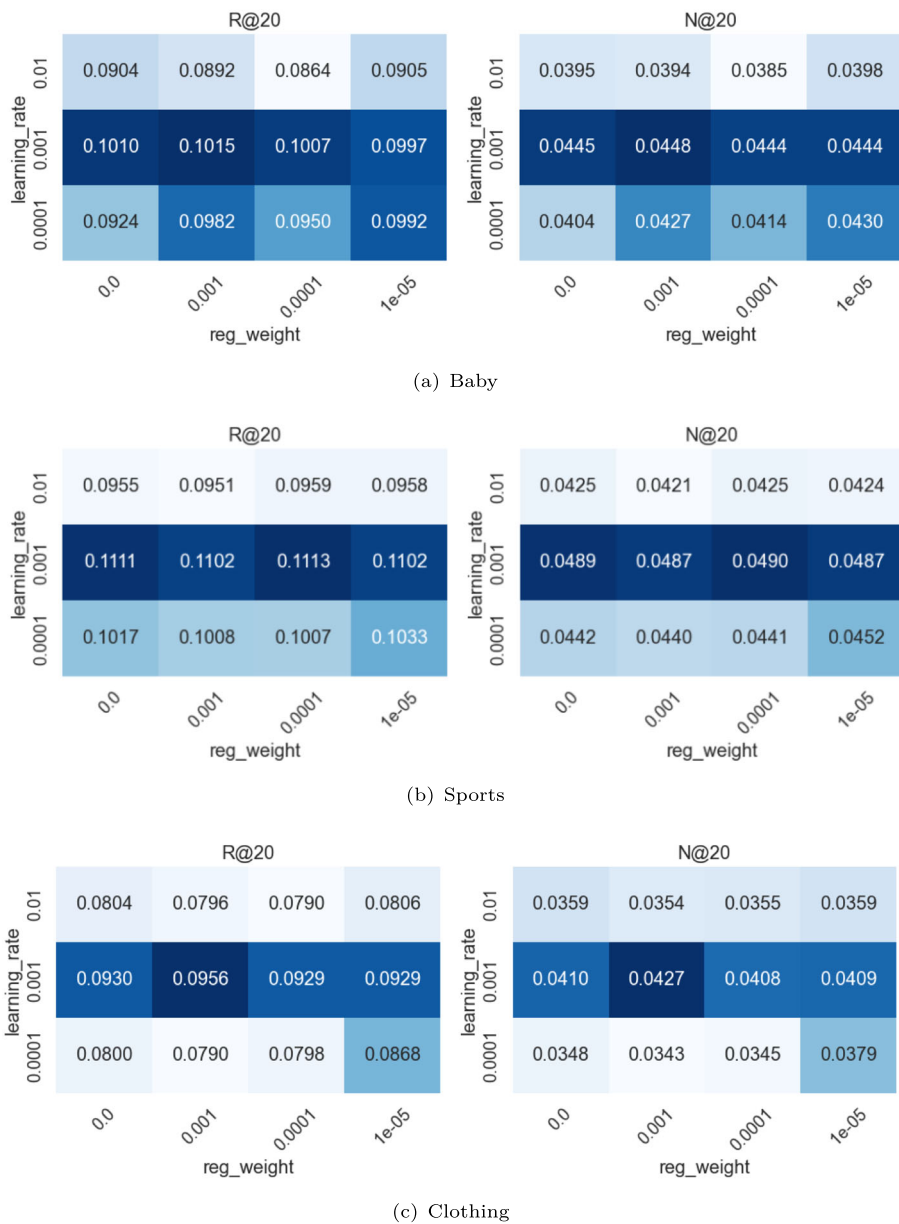


Fig. 9 Presents a sensitivity analysis of the hyperparameters for PMPR. Across various datasets, we explored the performance of PMPR with different loss weights λ , denoted as the reg_weight parameter on the x-axis, and with varying learning rates, referred to as Learning_rate

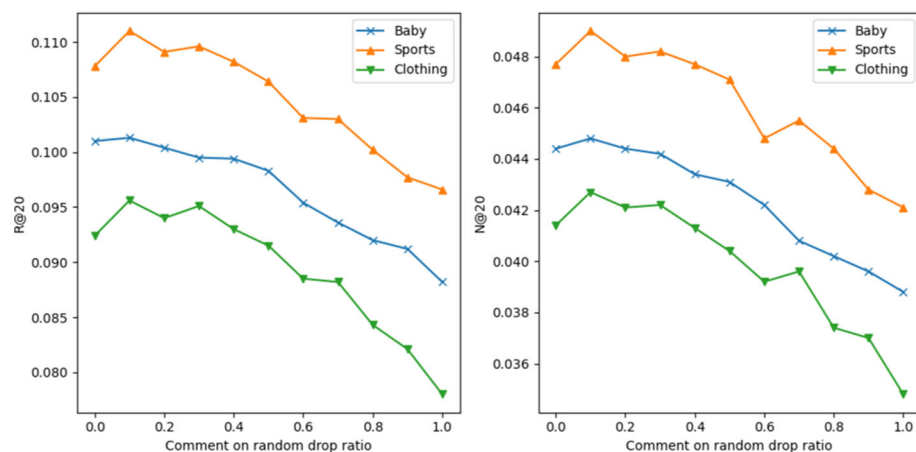


Fig. 10 The performance of PMPR varies with the proportion, ρ_c , of the item-item graph constructed based on review information

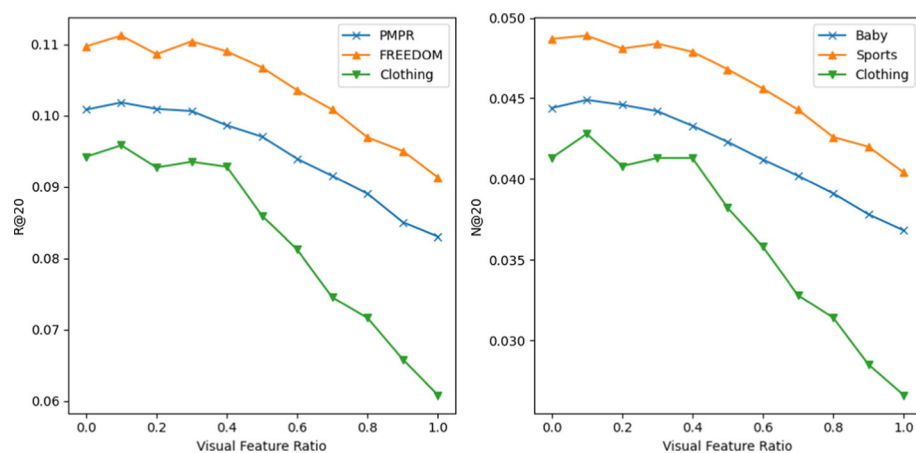


Fig. 11 The performance of PMPR varies with the fusion ratio of the item-item graph constructed from images and texts

6 Conclusion

This paper proposes an enhanced multimodal recommendation method that integrates reviews. Traditional multimodal graph learning methods rely solely on multimodal item data to learn user preferences for items. However, multimodal data often suffers from modality missing issues. To provide a more comprehensive overview of user preferences, we incorporate reviews to complement the missing modalities and enrich preference information related to items. We adopt a review-level modeling approach to process reviews and combine the item-side reviews with multimodal data to construct an item graph, thereby enhancing the representation of items. Simultaneously, we combine user-side reviews with the interaction matrix to learn users' preference information from reviews. Ultimately, the improved representations of users and items are utilized for the final recommendation. Empirical results demonstrate that our proposed model achieves significant effectiveness on three

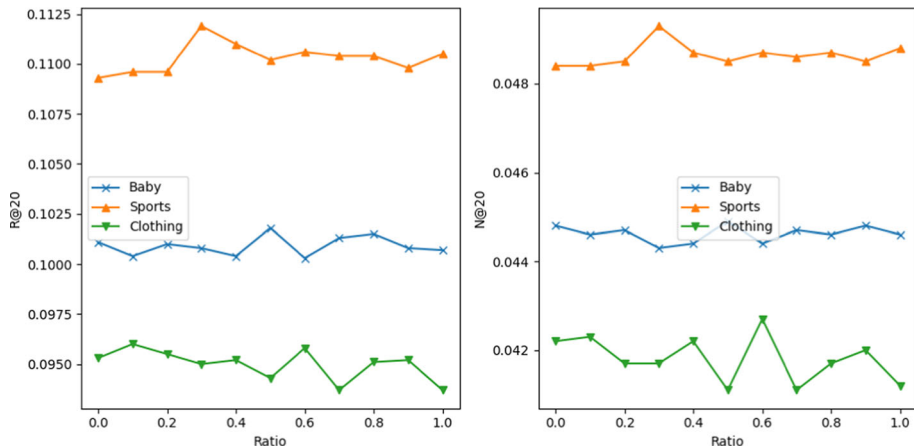


Fig. 12 The performance of PMPR varies with the fusion ratio of user features

public datasets, providing new insights and methodologies for further research and application in multimodal recommendation systems.

Author contributions F provided guidance to L and S, S helped L to survey the literature, and L was responsible for completing the experiment and writing the paper.

Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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